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Problem Statement:

Problem:

Emergency services, health officials, and researchers rely on disconnected datasets to understand stroke risks, patterns, and accessibility to health care across the country. Although the data exists, it is difficult to quickly identify which communities need prioritization and what the course of action needed is.

Design Aspect:

How can we present the information in a way that can:

1. Help decision makers quickly identify areas that need higher priority and take the necessary precautions.
2. Be easily interpretable for the general public.

Users:

User 1: Marisol Roach – Professional

Information:

Job: County Public Health Director

Works for: Jefferson County Department of Public Health

Responsibilities: Plan outreach events, Coordinate mobile clinics, Prioritize communities, Help plan allocation of limited public health funding.

Main goal: Determine which communities require intervention first.

Questions:

- Which counties have the greatest stroke burden?
- Which counties lack access to adequate care?
- What are initiatives that make sense? (given context)

Success: If Marisol leaves the platform knowing "County X should receive our next mobile stroke screening event."

User 2: Eddie Giles – General

Information:

Background: Retired mechanic who lives with his wife in New Haven, CT. He is 65 years old and has a medical history with type 2 diabetes and hypertension, placing him at an increased risk for stroke. His nearest hospital is approximately 15-25 minutes away. After learning about stroke prevention from his healthcare provider, Eddie wants to understand how New Haven County compares to neighboring counties and what healthcare resources are available if he experiences a stroke. Eddie is not familiar with public health terminology and benefits from plain-language explanations.

Main goal: Determine whether New Haven County is at higher risk for stroke and identify nearby healthcare resources and prevention opportunities.

Questions:

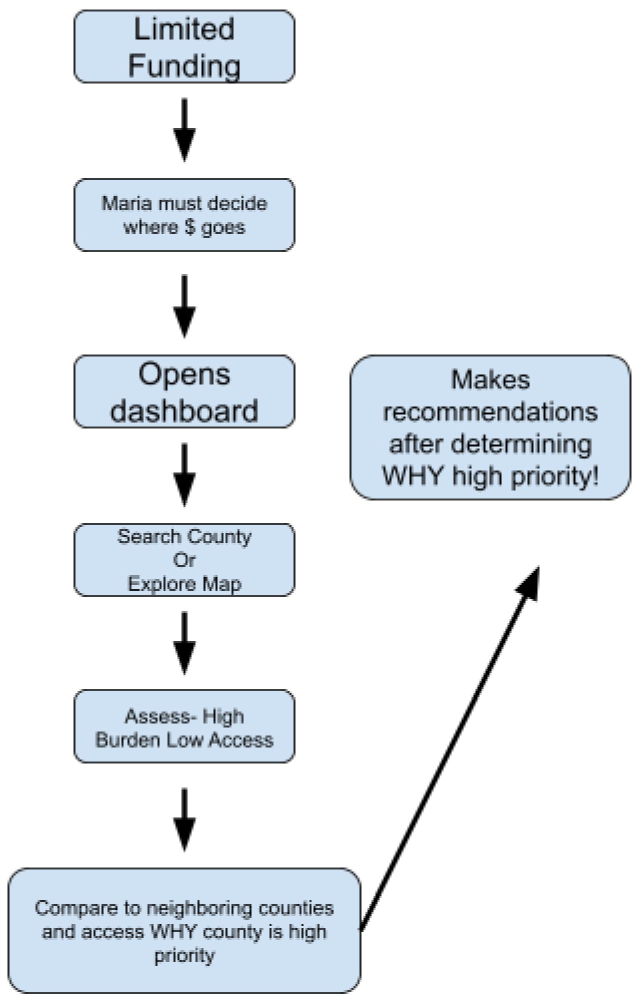
- Is New Haven County considered high risk for stroke?
- How does my county compare with neighboring counties?
- How far is the nearest stroke center or emergency department?
- What factors contribute to my county's stroke burden?
- What prevention resources or screening opportunities are available?

Success: If Eddie leaves the platform understanding his county's stroke risk, knowing where to seek care if needed, and feeling informed about prevention resources available in his community.

User Journey:

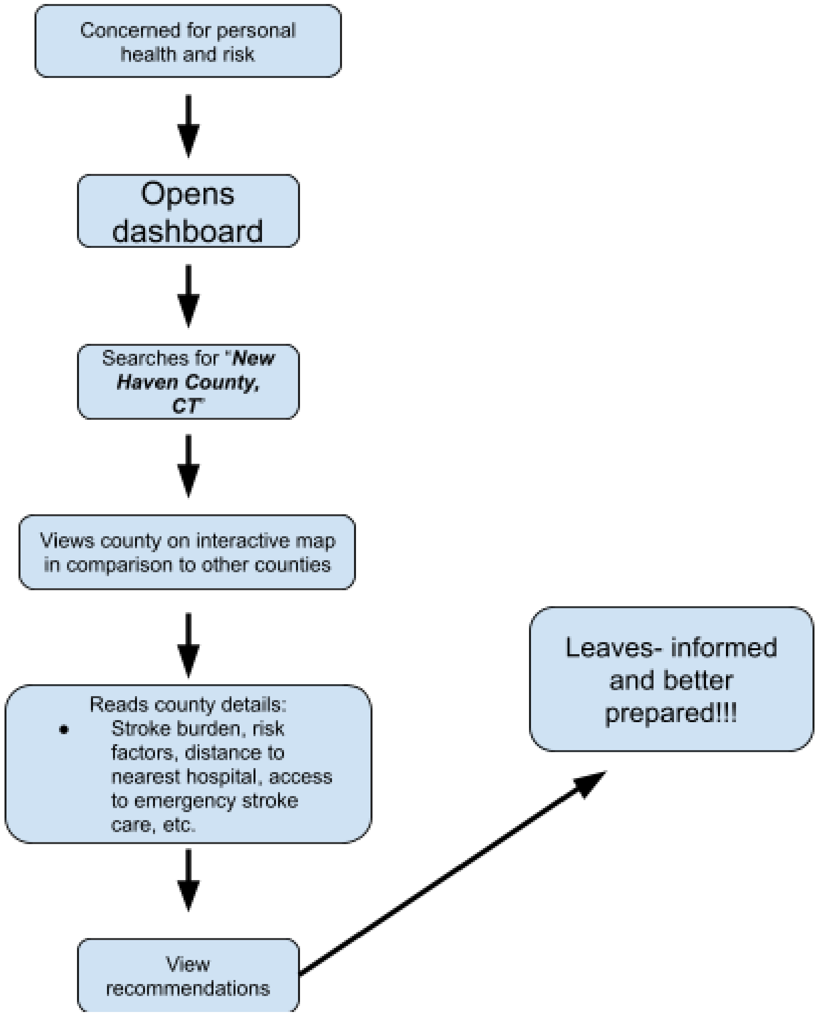
User 1: Marisol Roach

Journey:



User 2: Eddie Giles

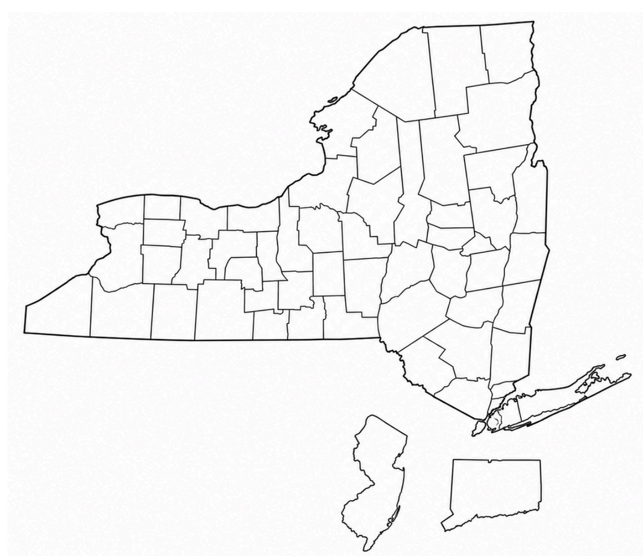
Journey:



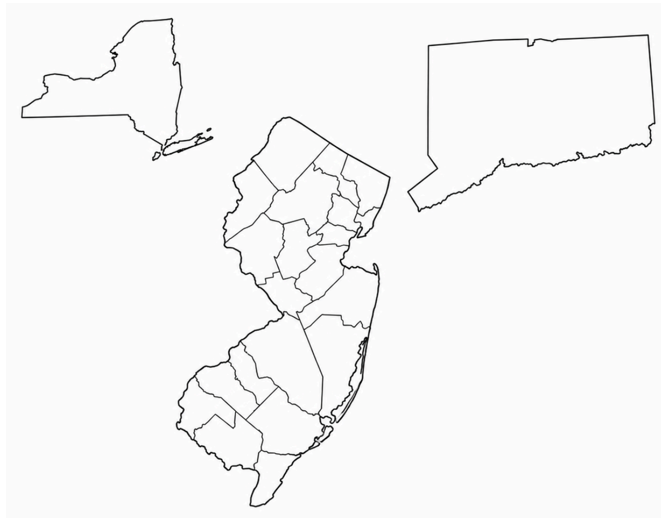
Design Opportunities/Research:

Idea 1: Help general users locate counties quickly

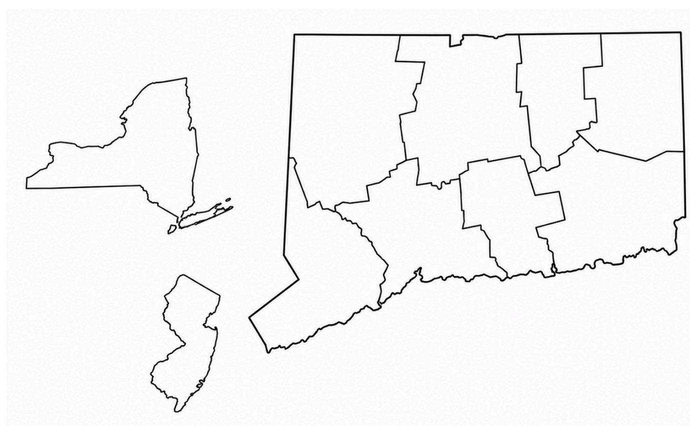
- Addresses: "Whether map and grid should be linked"
    - For overall use, both should be linked in some way, I would recommend through highlighting.
        - If the user clicks on a county in the map, it should highlight the grid section where the county is located (and vice versa).
            - This will take into account the user's Change Blindness.
                - Brings users attention to important information.
        - This would also allow for the map to remain clutterfree (Visual Attention). Vital information can be presented in a section besides the map and we can implement a zooming feature.
            - Establishes a clear visual hierarchy.
        - Eg. 1:



- Eg. 2:

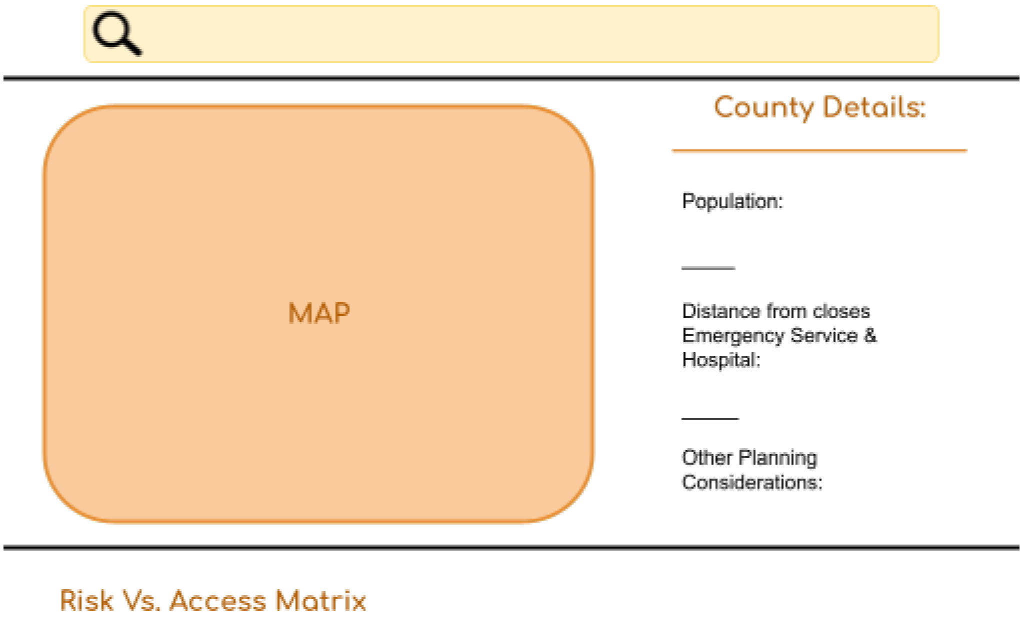


- Eg. 3:



- For more general users, it is highly likely that they will not know the exact location of their respective counties (which is likely the information they are looking for on the site). Users are more likely to know county names.
  - A search section for "*County Name:*" would help streamline the process.

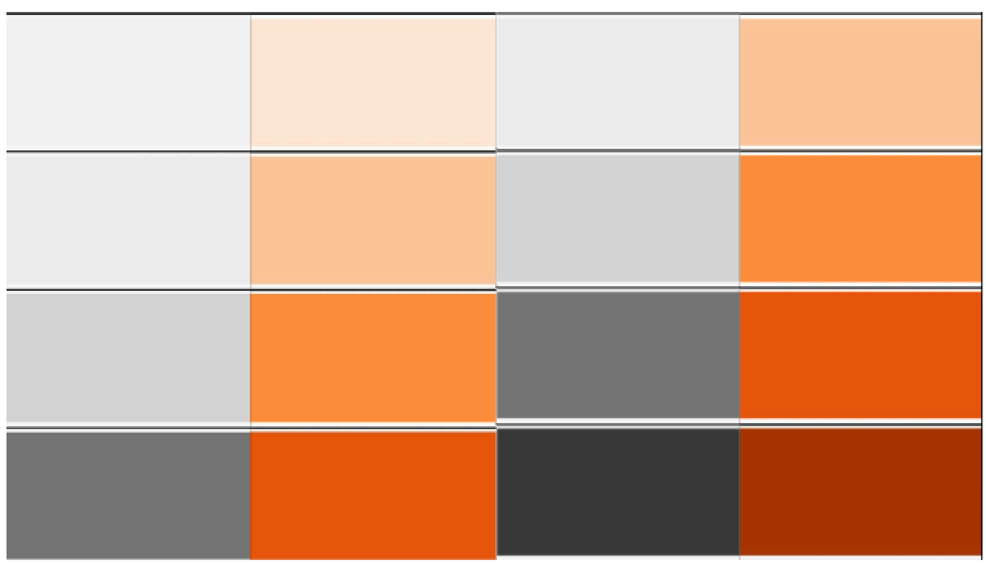
- Eg. of Setup:



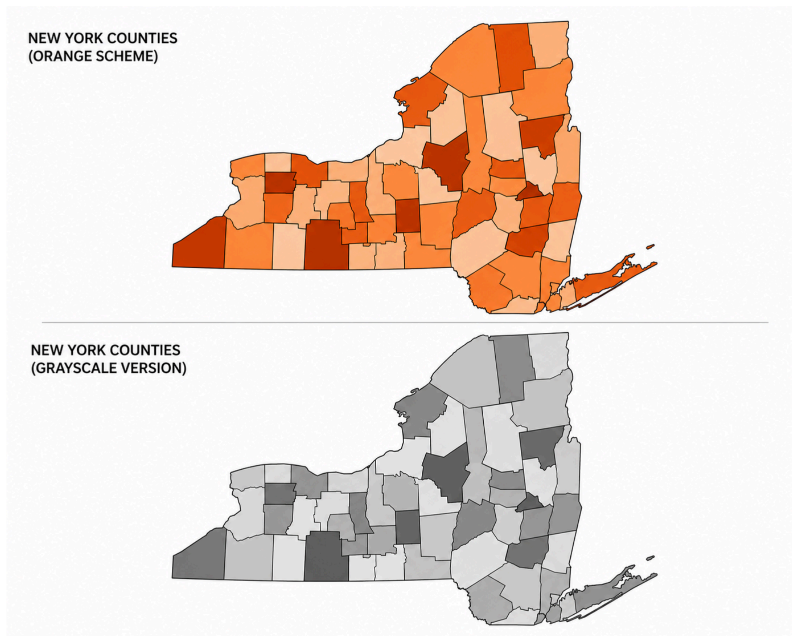
    - Related information is grouped together in logical order to aid in Information Processing.
- **Overall goal:** Helps users quickly identify high-priority counties, understand why they should be priorities, and support human decision making patterns effectively.

Idea 2: Color Palette

- Users perceive color differently depending on lighting conditions and individual differences in color vision. Because a portion of users experience color vision deficiencies, interfaces should not rely solely on hue (e.g., red vs. green) to communicate important information.
    - Use a sequential single-hue color palette (light → dark) instead of opposing colors like red and green.
    - Pair colors with labels, legends, or symbols so information remains understandable even if color perception differs.
    - Ensure sufficient contrast between adjacent color levels so users can distinguish counties based on both color and brightness.

- Color Blindness Research Specifically:
  - Cause:
    - Missing cones or abnormal cone function.
  - Monochromacy (Rod Monochromacy):
    - Characteristics:
      - No cones
      - Only rods
      - No color vision
      - Poor acuity
      - Sensitive to light
  - Dichromacy: One type of cone is missing.
    - Red-Green Deficiency
      - Missing M or L cones
    - Blue-Yellow Deficiency
      - Missing S cones (rare)
- Sequential Palette: Stroke burden "data [varies] from low to high values" therefore a sequential palette is most applicable for our design (PMC).
  - Take into account color context:
    - If using light colors against a light neutral background, utilize borders.
  - Orange: takes into account urgency without creating an emergency atmosphere.
    - 

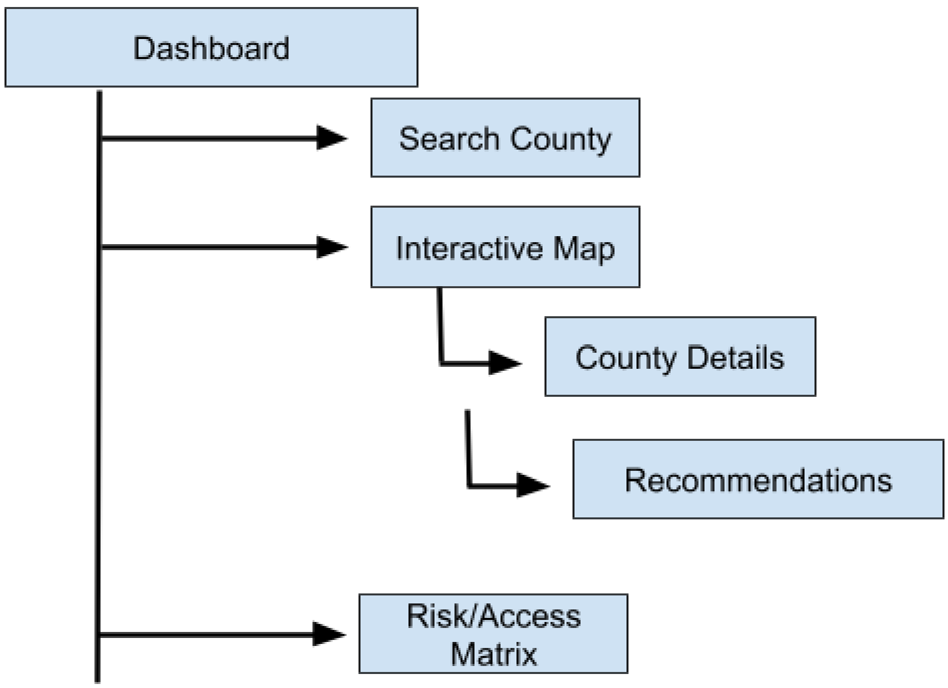
- Right side color scheme–
    - Stronger distinction between each category and clearer shifts in lightness scale for individuals who have reduced color discrimination.
    - Eg. New York Counties (randomized color assignments)



- The sequential orange palette communicates increasing stroke burden while avoiding the negative associations of red. Increasing luminance differences between categories also improves distinguishability for users with reduced color discrimination.

Information Architecture:

Sitemap:

- 

Information Architecture:

1: Visual Hierarchy

- Dashboards should present information according to importance.
  - Design Implications
    - Interactive county map serves as the primary focal point.
    - County details appear immediately adjacent to the selected county.
    - Supporting visualizations (Risk vs. Access Matrix) remains secondary.

2: Scanning Patterns

- Research suggests users typically scan pages using F-pattern or Z-pattern eye movements rather than reading every element sequentially. Important information should therefore appear early in the page layout.
  - Design Implications
    - Search bar positioned at the top of the page.
    - County map positioned in the upper-left region.
    - County details positioned beside the map.
    - Supporting charts and recommendations positioned below the primary content.

3: Information Grouping

- Related information should be grouped together to reduce cognitive load and improve information processing. Users should find information layout intuitive and simple.
  - Design Implications
    - Group each county's information together.
      - Stroke burden
      - Risk factors
      - Distance to nearest stroke center

        - Population
        - Planning considerations

4: Simplicity & Accessibility

- Dashboard should minimize unnecessary visual clutter while remaining accessible to users with varying levels of technical expertise and different visual abilities.
    - Design Implications
        - Minimize unnecessary interface elements.
        - Maintain sufficient contrast between adjacent colors.
        - Pair colors with labels and legends rather than relying on color alone.
        - Use plain-language explanations for general public users.

Sources:

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   A. Course Notes. Psychology of Perception (SJSU, Spring 2026). Topics referenced: opponent-process theory, color vision deficiencies, visual attention, and change blindness.

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   C. Bang Wong. Points of View: Color Blindness. Nature Methods. https://www.nature.com/articles/nmeth.1618

IV. Layout and Visual Hierarchy
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